DV1566 - Int. to Cloud Computing

Booklet for Laboratory 2

The purpose of Lab 2 is to practice the fundamental cloud mechanisms and resources that are:

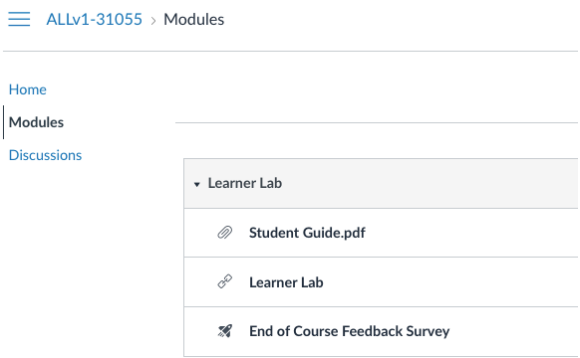
- Creation and use of VM[1], virtual storage, perimeters (Virtual Private Cloud – VPC and subnets)
- Monitoring of resource usage
- Virtual networking and security

Before starting the Lab, you are required to

- Accept the invitation to join the AWS Academy Learner Lab (ALLv2EN-US-LTI13-102285 (or 102286)
- Access the AWS Academy Learner Lab
- Read the Student Guide for the AWS Academy Learner Lab

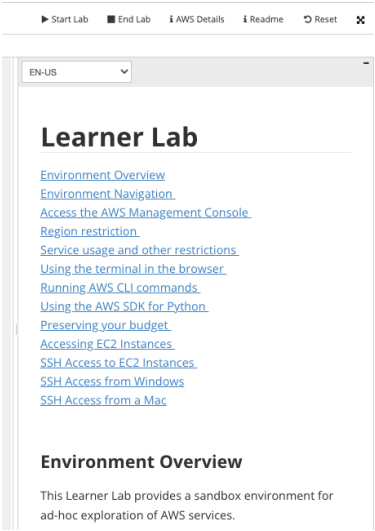
The AWS Academy Learner Lab offers a sandbox environment where you can run AWS resources. To start the Learner Lab environment:

- Click on Learner Lab to activate a persistent sandbox. After accepting terms and conditions you can start the lab session and open the AWS console.



[1] In the Amazon Web Service jargon, a virtual machine is called EC2 instance. In this document, I will use the terms virtual machine (VM) and EC2 instance interchangeably.

An overview of the services available in the AWS and restrictions is available under “Service usage and other restrictions” in the right frame



Laboratory 2 is composed of three tasks

1. Environment setup: setup a basic VPC and subnets, create and EC instance and create a new AMI
2. Monitoring: monitor the performance of an EC2 instance while under CPU and disk pressure.
3. Communication: connect two EC2 instance running in two subnets and monitor the performance while the instance is under web requests pressure.

Before starting the lab, I strongly suggest you: (i) read all the steps for each experience; (ii) figure out what are the goals and activities to be performed; (iii) study the documentation related to the tools you are going to use in the experience.

SUGGESTION: *each student has 100$ credit in his/her sandbox. The sandbox is used for the Lab 2 and for the final project. Hence: Use the price calculator before deploying your solution; monitor credit consumption, don’t forget to remove instantiated resources if not needed anymore. Note also that there are service that came for free (are included in the free tier), for example the t2.micro instances.*

Assessment (U/UX/G): You will be evaluated based on the submitted report. Specific/punctual instructions are not given to let you master the concepts and strengthen your critical thinking and problem-solving capability.

The lab is designed for Linux instances and, in particular, for Ubuntu image
However, you are free to choose other type of Linux AMI or Windows instances,

1 Task 1 – Environment Setup

The goal of this experience is to get familiar with: Amazon EC2 and related services: AWS storage options, security groups, and VPC. You will practice those concepts by creating and launching a VM on the Amazon cloud. This task consists of:

1. Create a public and private subnet in your default VPC and configure the routing table in a way that instances in the private subnet can communicate with the instance in the public subnet
2. Create a security group that allow SSH inbound connection
3. Create an EC2 instance in your public subnet and associate it the security group
4. Access your EC2 instance using and SSH client
5. Run an update (e.g. running sudo apt-get update)
6. Install the following software (if not yet installed with the OS) on the EC2 instance you launched: sysbench (check section 2.1 and 2.2), apache web server (apache httpd usually comes with ubuntu OS), and Apachebench (the apache2-utils, check section 3.1)
7. Captures a new AMI from the one you have modified in the previous steps. Let's call the new AMI *Lab2AMI,* it will be used in Task 2 and Task 3.

1.1 Resources

Useful resources for this task are

1. The following sections of the "Amazon EC2 User guide (for Linux instances)" https://aws.amazon.com/documentation/ec2/?icmpid=docs_menu:
   - What is amazon EC2
   - Setting UP
   - Getting Started
2. The user guide for Virtual Private Clouds (VPC), is available here http://docs.aws.amazon.com/AmazonVPC/latest/UserGuide/VPC_Introduction.html

2 Task 2 - Monitoring

The goal of this experience is to learn how to monitor VMs in a cloud environment. This task consists of

1. Enable EC2 instance monitoring for the instance you have launched in Task 1.

   To observe the CloudWatch statistics, access the CloudWatch service console.

2. Create your own dashboard with the metrics you wish to monitor. For example, you can create a dashboard with the following graphs: CPUutilization, DiskReadOps, DiskWriteOps, DiskReadBytes, DiskWriteBytes. Be aware that the t2.micro uses ESB storage, so you have to select CPU utilization from EC2 metrics and Disk related metrics from EBS metrics.
3. Generate workload for the EC2 isntance to practice with monitoring. Two types of workloads should be generated CPU and disk intensive.
    - CPU and disk intensive workload could be generated with tools like sysbench and stress-ng. Instructions to use sysbench follow in section 2.2. If you have previous experiences with other stress tools/benchmark tools, you can use those.

Tip. After you have launched the VM, I suggest updating the package database, e.g., running the `apt-get update`.

2.1 Resources

Useful resources for this task are

- The "Monitoring Amazon EC2" how-to guide. Focus on Automated and Manual monitoring, status check, detailed monitoring (1-minute period), ESB monitoring, CloudWatch, and available CloudWatch performance metrics. The guide is available at http://docs.aws.amazon.com/AWSEC2/latest/UserGuide/monitoring_ec2.html
- The CloudWatch user guide http://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/WhatIsCloudWatch.html
- The sysbench documentation; suggest links are

  https://github.com/akopytov/sysbench

  https://www.howtoforge.com/how-to-benchmark-your-system-cpu-file-io-mysql-with-sysbench

2.2 Examples of how to generate load with sysbench

First, you need to install sysbench. Access your EC2 instance and run the commands

```
sudo apt-get update
sudo apt-get install sysbench
```

Then, to check that sysbench works, you run the command

```
sudo sysbench --test=cpu --cpu-max-prime=200000 run
```

If everything works properly you can do a more intense CPU stress test. For example, the following command

```
for each in 1 2 4 8 16 32 64; do sysbench --test=cpu --cpu-max-prime=200000 --num-threads=$each run; done
```

runs sysbench 7 times and at each run, the number of threads is increased, starting from 1 up to 64.

The following steps are an example of how to stress the disk. First, you need to create a file set, and then you can perform read and write operations on the files. To create the file set, run the following command:

```
sysbench --test=fileio --file-total-size=8G --file-num=3 prepare
```

An example of a command to run the read and write tests 5 times, with an increasing number of threads (from 1 to 32), is the following:

```
for each in 1 4 8 16 32; do sysbench --test=fileio --file-total-size=8G
--file-test-mode=rndwr --max-time=240 --max-requests=0 --file-block-
size=4K --file-num=3 --num-threads=$each run; done;
```

Refer to the sysbench documentation and tutorials to understand the meaning of the command's options.

IMPORTANT: to observe a significant variation of the metric values, the amount of workload to be generated depends on the capacity of the EC2 instance you selected. Change the values of the benchmark's parameters given in the above examples according to the EC2 instance you selected.

3 Task 3 - Communication

The goal of this experience is to learn how to connect two EC2 instance running in two subnets and to monitor the network traffic with CloudWatch. This task consists of

1. Create a new EC2 instance configured with the Lab2AMI previously created, and name it WebServer. Launch the WebServer EC2 instance in the private subnet you created in Task 1. Now you should have the WebServer instance running in the private subnet and one EC2 instance running in the public subnet. Bot of them should run the Lab2AMI.
2. Check that from the EC2 instance in the public network you can send http requests to the WebServer in the private network.
3. Stress the WebServer in the private network with the Apachebench (ab) running in the public network. When running the test, monitor the network traffic on the web server (e.g., byte per second and number of active connections).

3.1 Resources

Useful resources for this task are:

- Amazon web service documentation on networking
  https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-networking.html
- Amazon web service documentation on security
  https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-security.html
- Apache web server documentation
  http://httpd.apache.org/
- ApacheBench documentation
  https://httpd.apache.org/docs/2.4/programs/ab.html

3.2 An example of ApacheBench stress test

ApacheBench (ab) is a benchmark that generates http requests. First, you should create a web page on the http server. Then, you can simulate multiple users requesting a web page with ab. The simplest syntax of the ab command is

```
ab -n <num_of_request> -c <num_of_concurrent_users> <your_web_site>
```

for example

```
ab -n 100000 -c 10 http://ip-10-251-50-12.ec2.internal/index.html
```

send 100000 http requests, generated from 50 concurrent users to the web server hosted on the VM with private IP address 10.251.50.12 and with internal hostname "ip-10-251-50-12.ec2.internal"

Note that the intensity of the load you generate depends on the value of the 'n' and 'c' parameters in the ab command. Play with the number of http requests (n) and the number of concurrent users (c) to observe significant values of the performance metrics.

4 Assessment

The group should produce a report integrated with screenshots describing the following:

1. The configuration of your security group and VPC. You should motivate your configuration choices.
2. The results of your performance test (Task 2 and Task 3). You should describe the following:
    - What are the performance metrics selected and why (motivate your choices)
    - How you generated the load (describe the scripts used and motivate your choices)
    - The meaning of your results (why you get such results and what they mean)

Always remember: to motivate your choices, to discuss the issues encountered and how you solved those issues.

No specific template for the report is provided. You can use the BTH template for reports or your own template. **On the first page of the template, please specify the group number and group members.**

The assessment is pass or fail (U/Ux/G). I expect that all the group members will contribute to the work equally. However, do not hesitate to contact me if group members are not contributing.